

A Sparse Reconstruction Framework for Multi-Transmitter Radio Map Estimation

Bilginer Oral[†], Matthew Dwyer[‡], Kevin S. Chan[‡], and Erdem Koyuncu[†]

[†]Department of Electrical and Computer Engineering, University of Illinois Chicago, Chicago, IL, USA

[‡]DEVCOM U.S. Army Research Laboratory, Adelphi, MD, USA

Emails: {boral, ekoyuncu}@uic.edu, {matthew.r.dwyer7.civ, kevin.s.chan.civ}@army.mil

Abstract—Radio map estimation has drawn increasing attention because of its broad utility in wireless communications and military operations. We focus on the *sampling-based* setting, where the target radio maps are estimated from sparse observations without access to the underlying transmitter realizations. We formulate this problem as a sparse reconstruction task for radio maps drawn from an underlying distribution. Under regularity assumptions, we show a decomposition of the expected population risk into an approximation term that scales as $O(s^{-2/d})$ with the number of observed samples and an estimation term that depends on the number of training examples. We then empirically study how practical error rates compare with the scaling behavior predicted by the theory. Finally, we introduce RadioET, the first energy-based reconstruction method for radio map estimation. Experiments on a multi-transmitter radio map benchmark demonstrate the effectiveness of RadioET.

Index Terms—Radio map estimation, sparse reconstruction, energy-based models.

I. INTRODUCTION

Radio map estimation (RME) is the process of reconstructing quantities such as path loss or received signal strength across a geographic area from measurements at known locations [1]. Radio maps support network planning, resource allocation, and situational awareness [2]. Such applications are essential in emerging 6G and defense settings. Thus, it is critical to develop robust RME methods that generalize across diverse propagation environments.

We consider two RME scenarios depending on whether the transmitter (TX) realization is known [2]. In the *TX-conditioned* setting, the transmitter realization (the number of transmitters, their locations, and transmit powers) is modeled as part of the input. Consequently, the learning task reduces to approximating a deterministic function of the environment and transmitter parameters. In this paper, we focus on the *sampling-based* setting [3], where the transmitter realization is unknown. Unlike prior work that treats antenna locations as design variables to optimize system performance [4], [5], the transmitter geometry must be inferred indirectly from sparse measurements. In this case, the input consists of environment information and a sparse set of measurements.

In this work, we view this sparse reconstruction task through the lens of standard approximation and estimation error rates. Our contributions are summarized as follows: leftmargin=*

- We formulate sampling-based *multi-transmitter* radio map estimation, where the transmitter count, locations,

and powers are unknown and must be inferred indirectly from sparse measurements. We also construct a multi-transmitter benchmark from single-transmitter RadioMapSeer maps by superposing fields in power space.

- We develop an approximation–estimation analysis for sparse RME and derive an asymptotic reconstruction-risk bound with an $O(s^{-2/d})$ approximation term in the number of observed samples s over a d -dimensional domain, and a finite-sample estimation term that decays as $O((\frac{\log n}{n})^{\frac{1}{2}})$ with the number of training maps n . This provides a theoretical sparsity-scaling reference beyond the primarily numerical focus of most learning-based RME studies.
- We introduce RadioET, to our knowledge the first energy-based architecture for RME, and show that it consistently outperforms existing models on the proposed multi-transmitter benchmark, with empirical sparsity scaling consistent with the $O(1/s)$ reference rate in two dimensions.

The rest of this paper is organized as follows. Section II reviews related work on RME approaches. Sections III and IV formalize the problem and present the theoretical analysis. Section V describes the architecture of RadioET. Section VI presents experimental results on the multi-transmitter RME benchmark. Finally, Section VII concludes the paper with a discussion of limitations and future directions.

II. RELATED WORK

Early data-driven RME approaches often used spatial interpolation and geostatistical methods, such as Kriging [1] and inverse distance weighting (IDW) [6]. These methods reconstruct the radio field from sparse observations by assuming spatial correlation or smooth distance-based variation. However, they typically do not explicitly model environmental structure or propagation physics, which can limit accuracy in obstructed and complex settings.

Deep learning has recently become a common approach to RME. In the TX-conditioned scenario, representative architectures include convolutional encoder-decoder models and generative methods such as RadioUNet [7], RadioMamba [8], RadioDiff [9], PhyRMDM [10] and RME-GAN [11]. Joint reconstruction has also been explored, where UAV measurements are used to estimate both the radio and environment maps [12]. Related work has investigated the sampling-based

RME problem under extreme levels of sparsity. For example, DAT-Unet [3] uses deformable attention with neighborhood attention to efficiently process sparse measurements. A physics-inspired approach to RME can be found in [13].

Beyond RME, sparse reconstruction has been studied in other domains. Recent work includes REPA [14] for reconstructing sparsely observed vector fields using self-supervised learning, and the Energy Transformer [15], [16] for reconstructing flow fields through energy minimization. While these methods demonstrate the effectiveness of learned priors for sparse reconstruction, energy-based approaches have not yet been explored for radio map estimation.

III. PROBLEM FORMULATION

A. Preliminaries

Let $D = [0, 1]^d \subset \mathbb{R}^d$ denote the spatial domain, taken to be the unit cube so that its volume is $|D| = 1$. Let $\mathcal{B}$ denote the environmental information available to the estimator (e.g. buildings, cars). A transmitter configuration is denoted by $C = (m, \{(t_j, p_j)\}_{j=1}^m)$, where m is the number of active transmitters, $t_j \in D$ is the location of transmitter j , and $p_j > 0$ is its linear transmit-power scale.

Let g be a deterministic radio wave propagation operator from transmitter configurations and environment maps to radio maps. In the TX-conditioned setting, C and $\mathcal{B}$ are both part of the input, and the conditional field law is the Dirac mass $\mathcal{P}_{F|C, \mathcal{B}}(\cdot | c, b) = \delta_{g(c, b)}$, where F denotes a random map. In other words, conditioned on the full input, there is no non-degenerate distribution over target fields and the learning problem reduces to deterministic operator regression.

In the sampling-based setting, the transmitter configuration is unobserved. The estimator is provided with the environment $\mathcal{B}$, but not the configuration C . A random configuration C and environment $\mathcal{B}$ are drawn and passed through the propagation operator to generate a radio map $F = g(C, \mathcal{B})$. We denote by $F \sim \mathcal{P}$ the resulting distribution over radio maps induced by the randomness in C and $\mathcal{B}$. Since C is not observed by the estimator, the target field remains stochastic even after conditioning on the available inputs.

B. Teacher–student formalization

The *teacher* randomly samples maps from the distribution $\mathcal{P}$. Let F denote a random map: $F : D \rightarrow \mathbb{R}^k, F \sim \mathcal{P}$, where $k = 1$ for scalar radio maps. For a fixed sparsity level s , observation locations $X_1, \dots, X_s$ are drawn independently and uniformly in D , and the sparse observation tuple is

$$O_s = ((X_1, Y_1), \dots, (X_s, Y_s)),$$

where $Y_j = F(X_j)$ is the observed value at location X_j . A *student* reconstructor with parameters $\theta \in \Theta$ maps the environment information and observations to a full field

$$h_\theta(\mathcal{B}, O_s) : D \rightarrow \mathbb{R}^k.$$

We write $\widehat{F}_\theta(x; \mathcal{B}, O_s) = h_\theta(\mathcal{B}, O_s)(x)$ for its value at a point x . For two fields $f, \tilde{f} : D \rightarrow \mathbb{R}^k$, the reconstruction loss is the mean squared error

$$L(f, \tilde{f}) = \int_D \|f(x) - \tilde{f}(x)\|_2^2 dx.$$

The population risk of a reconstructor h_θ is

$$\mathcal{L}_{\text{gen}}^{(s)}(h_\theta) = \mathbb{E}[L(F, h_\theta(\mathcal{B}, O_s))],$$

and the best-in-class risk is

$$\delta(s) = \inf_{\theta \in \Theta} \mathcal{L}_{\text{gen}}^{(s)}(h_\theta).$$

Given n training examples $T_s^{(i)} = (\mathcal{B}^{(i)}, O_s^{(i)}, F^{(i)})$, the empirical risk is

$$\mathcal{L}_n^{(s)}(h_\theta) = \frac{1}{n} \sum_{i=1}^n L(F^{(i)}, h_\theta(\mathcal{B}^{(i)}, O_s^{(i)})),$$

and an empirical risk minimizer is $h_{\theta_{n,s}^*}$, where

$$\theta_{n,s}^* \in \arg \min_{\theta \in \Theta} \mathcal{L}_n^{(s)}(h_\theta).$$

The expected population risk of the ERM is

$$\delta(n, s) = \mathbb{E}[\mathcal{L}_{\text{gen}}^{(s)}(h_{\theta_{n,s}^*})],$$

where the outer expectation is over the random training set.

IV. THEORETICAL ANALYSIS

In this section, we state and prove the sample-complexity consequences of the sparse-field formalization. While the main arguments follow from [17] and [18], we find it valuable to present the results compactly and in a self-contained manner.

Throughout this section we work in a generic setting whose estimator input is the sparse observation tuple O_s . To keep the notation light, we suppress the environment $\mathcal{B}$ of Section III and absorb it into the induced field law $\mathcal{P}$. Accordingly we drop $\mathcal{B}$ wherever it occurs, i.e. write $h_\theta(O_s)$ for $h_\theta(\mathcal{B}, O_s)$ and a training example becomes $T_s = (O_s, F)$. We emphasize that the results of this section rely on the assumptions below:

- 1) The domain $D = [0, 1]^d$ is the unit cube, with volume $|D| = 1$.
- 2) For each fixed s , the sample locations $X_1, \dots, X_s$ are i.i.d. uniform on D .
- 3) Every field drawn by the teacher is L_0 -Lipschitz.
- 4) Teacher and student fields are uniformly bounded, i.e. $\|F(x)\|_2 \leq B$ and $\|h_\theta(O_s)(x)\|_2 \leq B$ for all $x \in D$ and all $\theta \in \Theta$.
- 5) For fixed s , the induced loss class

$$\mathcal{G}_s = \{T_s \mapsto L(F, h_\theta(O_s)) : \theta \in \Theta\}$$

has finite pseudo-dimension d_s .

The proof of the approximation bound proceeds in two stages. First, we show that a nearest-neighbor reconstructor achieves population risk $\mathcal{O}(s^{-2/d})$. Next, we show that neural architectures with parameter class Θ can approximate the nearest-neighbor reconstructor arbitrarily well in mean square, and that the same rate transfers to $\delta(s)$.

Define the *nearest-neighbor (NN) reconstructor* h_{NN} , which predicts at a point x the value observed at the closest sample point,

$$\hat{F}_{\text{NN}}(x; O_s) = h_{\text{NN}}(O_s)(x) = Y_{i(x)},$$

where $i(x) \in \arg \min_{1 \leq j \leq s} \|x - X_j\|_2$. Writing $\mathbf{X}_s = (X_1, \dots, X_s)$ for the sample locations, the distance from x to its nearest sample point is

$$\text{dist}(x, \mathbf{X}_s) \triangleq \min_{1 \leq j \leq s} \|x - X_j\|_2.$$

Lemma 1 (Lipschitz bound) *The population risk of the NN reconstructor satisfies*

$$\mathcal{L}_{\text{gen}}^{(s)}(h_{\text{NN}}) \leq L_0^2 \int_D \mathbb{E}[\text{dist}(x, \mathbf{X}_s)^2] dx.$$

Proof: By Lipschitz continuity (Assumption 3), for every $x \in D$,

$$\begin{aligned} \|F(x) - \hat{F}_{\text{NN}}(x; O_s)\|_2 &= \|F(x) - F(X_{i(x)})\|_2 \leq L_0 \|x - X_{i(x)}\|_2 \\ &= L_0 \text{dist}(x, \mathbf{X}_s). \end{aligned}$$

Squaring and integrating over D ,

$$\begin{aligned} \int_D \|F(x) - \hat{F}_{\text{NN}}(x; O_s)\|_2^2 dx &\leq L_0^2 \int_D \text{dist}(x, \mathbf{X}_s)^2 dx. \end{aligned}$$

Taking expectations and swapping with the integral over D (Tonelli's theorem),

$$\begin{aligned} \mathcal{L}_{\text{gen}}^{(s)}(h_{\text{NN}}) &\leq L_0^2 \mathbb{E} \left[\int_D \text{dist}(x, \mathbf{X}_s)^2 dx \right] \\ &= L_0^2 \int_D \mathbb{E}[\text{dist}(x, \mathbf{X}_s)^2] dx. \end{aligned}$$

Lemma 2 (Geometric bound on NN distance) *There exists a constant $C_1 > 0$ depending only on d [17, Theorem 2.1], such that*

$$\mathbb{E}[\text{dist}(x, \mathbf{X}_s)^2] \leq C_1 s^{-2/d} \quad \text{for all } s \geq 1, x \in D.$$

Proof: Fix $x \in D$ and write $R \triangleq \text{dist}(x, \mathbf{X}_s) = \min_i \|x - X_i\|_2$. Since the X_i are i.i.d. uniform on the unit cube D (Assumption 2), a single sample lands within distance r of x with probability $q(r) = |B(x, r) \cap D|$, and by independence $\mathbb{P}(R \geq r) = (1 - q(r))^s$. Using $\mathbb{E}[R^2] = \int_0^\infty \mathbb{P}(R^2 \geq t) dt$ and substituting $t = r^2$,

$$\mathbb{E}[R^2] = \int_0^\infty 2r (1 - q(r))^s dr.$$

Ignoring boundary effects, a ball of radius r has volume $|B(x, r)| = \kappa_d r^d$, where κ_d is the volume of the unit ball, so $q(r) \approx \kappa_d r^d$. Substituting $v = \kappa_d r^d$, for which $2r dr = \frac{2}{d} \kappa_d^{-2/d} v^{2/d-1} dv$,

$$\mathbb{E}[R^2] \approx \frac{2}{d} \kappa_d^{-2/d} \int_0^1 v^{2/d-1} (1-v)^s dv.$$

Using $1 - v \leq e^{-v}$ and then substituting $w = sv$,

$$\begin{aligned} \int_0^1 v^{2/d-1} (1-v)^s dv &\leq \int_0^\infty v^{2/d-1} e^{-sv} dv \\ &= s^{-2/d} \int_0^\infty w^{2/d-1} e^{-w} dw. \end{aligned}$$

The last integral is the Euler gamma function $\Gamma(2/d)$ (finite since $2/d > 0$). Hence $\mathbb{E}[R^2] \leq C_1 s^{-2/d}$ with C_1 depending only on d . ■

Corollary 1 (Rate for the NN reconstructor) *Combining Lemmas 1 and 2, there exists a constant $C_{\text{NN}} = L_0^2 C_1 |D| > 0$ such that for all $s \geq 1$,*

$$\mathcal{L}_{\text{gen}}^{(s)}(h_{\text{NN}}) \leq L_0^2 \int_D C_1 s^{-2/d} dx = C_{\text{NN}} s^{-2/d}.$$

Note that $s^{-2/d}$ decay in Corollary 1 faithfully reproduces the classical rate for Lipschitz functions [17, Theorem 14.5]. Specifically, since we are in the noiseless regime, the variance term $\sigma^2 = 0$, and with $k = 1$ neighbor the weight vector collapses to $v_{n1} = 1$ and $v_{ni} = 0$ for $i \geq 2$. Thus only the leading term in the bias sum survives. The case $d \geq 2$ of Theorem 14.5 then reduces to $c'_d L^2 (1/n)^{2/d}$, which matches our $C_{\text{NN}} s^{-2/d}$ under the identifications $n \leftrightarrow s$, $L \leftrightarrow L_0$, and $C_1 = c'_d$ (using $|D| = 1$).

We now introduce a joint input for the random query point and the observations, $Z_s \triangleq (X, X_1, \dots, X_s, Y_1, \dots, Y_s)$ with $X \sim \text{Unif}(D)$, so that $\hat{F}_\theta(Z_s) = \hat{F}_\theta(X; O_s)$ and $\hat{F}_{\text{NN}}(Z_s) = \hat{F}_{\text{NN}}(X; O_s)$. With this notation the population risk becomes

$$\mathcal{L}_{\text{gen}}^{(s)}(h_\theta) = |D| \mathbb{E}[\|F(X) - \hat{F}_\theta(Z_s)\|_2^2]. \quad (1)$$

Throughout the rest of this section we quantify how well one reconstructor approximates another in mean square over Z_s , i.e. by expectations of the form $\mathbb{E}\|g(Z_s)\|_2^2$.

Lemma 3 (Risk continuity) *Let h_1, h_2 be two reconstructors and define $e(Z_s) \triangleq \hat{F}_1(Z_s) - \hat{F}_2(Z_s)$ for the difference of their predictions on the random input Z_s . Then, under Assumption 4,*

$$\begin{aligned} |\mathcal{L}_{\text{gen}}^{(s)}(h_1) - \mathcal{L}_{\text{gen}}^{(s)}(h_2)| &\leq |D| \left(4B (\mathbb{E}\|e(Z_s)\|_2^2)^{1/2} + \mathbb{E}\|e(Z_s)\|_2^2 \right). \end{aligned}$$

Proof: Write the residuals at the random input, $r_i \triangleq F(X) - \hat{F}_i(Z_s)$ for $i = 1, 2$, so that $r_1 = r_2 - e$. By the population-risk identity (1),

$$\begin{aligned} \mathcal{L}_{\text{gen}}^{(s)}(h_1) - \mathcal{L}_{\text{gen}}^{(s)}(h_2) &= |D| \mathbb{E}[\|r_1\|_2^2 - \|r_2\|_2^2] = |D| \mathbb{E}[-2\langle r_2, e \rangle + \|e\|_2^2]. \end{aligned}$$

Taking absolute values and applying Cauchy-Schwarz,

$$\begin{aligned} |\mathcal{L}_{\text{gen}}^{(s)}(h_1) - \mathcal{L}_{\text{gen}}^{(s)}(h_2)| &\leq 2|D| (\mathbb{E}\|r_2\|_2^2)^{1/2} (\mathbb{E}\|e\|_2^2)^{1/2} + |D| \mathbb{E}\|e\|_2^2. \end{aligned}$$

Since $\|r_2\|_2 = \|F(X) - \hat{F}_2(Z_s)\|_2 \leq 2B$ by Assumption 4, we have $(\mathbb{E}\|r_2\|_2^2)^{1/2} \leq 2B$, and therefore

$$\begin{aligned} & |\mathcal{L}_{\text{gen}}^{(s)}(h_1) - \mathcal{L}_{\text{gen}}^{(s)}(h_2)| \\ & \leq |D| \left(4B (\mathbb{E}\|e\|_2^2)^{1/2} + \mathbb{E}\|e\|_2^2 \right). \end{aligned}$$

Proposition 1 (NN approximation) *Suppose that for parameters $\theta \in \Theta$ there exists a reconstructor of the form*

$$\hat{F}_\theta(x; o_s) = \rho_\theta \left(x, \sum_{j=1}^s a_{\theta,j}(x; x_1, \dots, x_s) v_\theta(x_j, y_j) \right).$$

Then such a reconstructor can approximate the NN reconstructor arbitrarily well in mean square, i.e. for every $\varepsilon > 0$ there exists a $\theta \in \Theta$ with

$$(\mathbb{E}\|\hat{F}_\theta(Z_s) - \hat{F}_{\text{NN}}(Z_s)\|_2^2)^{1/2} \leq \varepsilon.$$

Proof: We give a proof by construction. Assume $\beta \in \Theta$ and consider the Nadaraya-Watson interpolant of [17, Section 10.2]

$$\begin{aligned} a_{\beta,j}(x; x_1, \dots, x_s) &= \frac{e^{-\beta\|x-x_j\|_2^2}}{\sum_{\ell=1}^s e^{-\beta\|x-x_\ell\|_2^2}}, \\ v_\beta(x_j, y_j) &= y_j, \quad \rho_\beta(x, u) = u, \end{aligned}$$

so that the reconstructor is

$$\hat{F}_\beta(x; o_s) = \sum_{j=1}^s \frac{e^{-\beta\|x-x_j\|_2^2}}{\sum_{\ell=1}^s e^{-\beta\|x-x_\ell\|_2^2}} y_j.$$

We show that $\mathbb{E}\|\hat{F}_\beta(Z_s) - \hat{F}_{\text{NN}}(Z_s)\|_2^2 \rightarrow 0$ as $\beta \rightarrow \infty$.

The samples are distinct with probability one because each X_i has a density, so $\mathbb{P}(X_i = X_j) = 0$ for $i \neq j$. Thus, assume distinct sample locations $x_1, \dots, x_s$. We consider two cases for a random query point X : either X has a unique nearest sample, or it is equidistant from two or more samples (a tie). X is equidistant from x_i and x_j only when $\|X - x_i\|_2 = \|X - x_j\|_2$. This forces X onto the hyperplane $\{z : 2z \cdot (x_j - x_i) = \|x_j\|_2^2 - \|x_i\|_2^2\}$. There are at most $\binom{s}{2}$ such hyperplanes, and X lands on their union with probability zero. Hence, X has a unique nearest sample $x_{i(x)}$ with probability one.

Now fix (x, o_s) with a unique nearest sample $x_{i(x)}$, and for $j \neq i(x)$ set $\Delta_j = \|x - x_j\|_2^2 - \|x - x_{i(x)}\|_2^2 > 0$. Dividing the numerator and denominator of the nearest sample's softmax weight by $e^{-\beta\|x - x_{i(x)}\|_2^2}$,

$$a_{\beta,i(x)}(x; x_1, \dots, x_s) = \frac{1}{1 + \sum_{j \neq i(x)} e^{-\beta\Delta_j}} \xrightarrow{\beta \rightarrow \infty} 1,$$

since each $\Delta_j > 0$, and the remaining weights tend to 0. Therefore, pointwise,

$$\begin{aligned} \hat{F}_\beta(x; o_s) &= \sum_{j=1}^s a_{\beta,j}(x; x_1, \dots, x_s) y_j \\ &\longrightarrow y_{i(x)} = \hat{F}_{\text{NN}}(x; o_s). \end{aligned}$$

The softmax weights are nonnegative and sum to one, and $\|y_j\|_2 \leq B$ by Assumption 4, so $\|\hat{F}_\beta(Z_s)\|_2 \leq B$ and $\|\hat{F}_{\text{NN}}(Z_s)\|_2 \leq B$. Thus $\|\hat{F}_\beta(Z_s) - \hat{F}_{\text{NN}}(Z_s)\|_2^2 \leq 4B^2$. This

integrand is bounded by the constant $4B^2$ and tends to 0 with probability one, so by dominated convergence

$$\mathbb{E}\|\hat{F}_\beta(Z_s) - \hat{F}_{\text{NN}}(Z_s)\|_2^2 \longrightarrow 0 \quad \text{as } \beta \rightarrow \infty.$$

In particular, given $\varepsilon > 0$, choosing β so that $\mathbb{E}\|\hat{F}_\beta(Z_s) - \hat{F}_{\text{NN}}(Z_s)\|_2^2 \leq \varepsilon^2$ yields a reconstructor with root-mean-square error at most ε . $\blacksquare$

Theorem 1 (Approximation error) *Assume Θ contains the parameter β of Proposition 1. Then there exists a constant $C > 0$, independent of s , such that*

$$\delta(s) \leq C s^{-2/d}. \quad (2)$$

Consequently, $\delta(s) \rightarrow 0$ as $s \rightarrow \infty$.

Proof: By Corollary 1, there exists a constant $C_{\text{NN}} > 0$, independent of s , such that $\mathcal{L}_{\text{gen}}^{(s)}(h_{\text{NN}}) \leq C_{\text{NN}} s^{-2/d}$. Fix $s \in \mathbb{N}$. By Proposition 1, for every $\varepsilon > 0$ there is a $\theta_\varepsilon \in \Theta$ with $(\mathbb{E}\|\hat{F}_{\theta_\varepsilon}(Z_s) - \hat{F}_{\text{NN}}(Z_s)\|_2^2)^{1/2} \leq \varepsilon$. Applying Lemma 3 with $h_1 = h_{\theta_\varepsilon}$ and $h_2 = h_{\text{NN}}$ gives

$$\mathcal{L}_{\text{gen}}^{(s)}(h_{\theta_\varepsilon}) \leq \mathcal{L}_{\text{gen}}^{(s)}(h_{\text{NN}}) + |D|(4B\varepsilon + \varepsilon^2).$$

Therefore

$$\begin{aligned} \delta(s) &= \inf_{\theta \in \Theta} \mathcal{L}_{\text{gen}}^{(s)}(h_\theta) \leq \mathcal{L}_{\text{gen}}^{(s)}(h_{\theta_\varepsilon}) \\ &\leq C_{\text{NN}} s^{-2/d} + |D|(4B\varepsilon + \varepsilon^2). \end{aligned}$$

Since this holds for every $\varepsilon > 0$, letting $\varepsilon \downarrow 0$ yields $\delta(s) \leq C_{\text{NN}} s^{-2/d}$, and setting $C = C_{\text{NN}}$ proves the claim. $\blacksquare$

We now study the generalization of the ERM as the number of independent training examples n grows. Specifically, we show that $\delta(n, s) - \delta(s) \rightarrow 0$ as $n \rightarrow \infty$. Recall that a training example is $T_s = (O_s, F)$, where O_s is the observation tuple and F the teacher field. The parameter space Θ induces the loss class $\mathcal{G}_s = \{g_\theta : \theta \in \Theta\}$ of Assumption 5, where $g_\theta(T_s) \triangleq L(F, h_\theta(O_s))$ and $\text{Pdim}(\mathcal{G}_s) = d_s < \infty$. Accordingly, the population and empirical risks read

$$\mathcal{L}_{\text{gen}}^{(s)}(h_\theta) = \mathbb{E}[g_\theta(T_s)], \quad \mathcal{L}_n^{(s)}(h_\theta) = \frac{1}{n} \sum_{i=1}^n g_\theta(T_s^{(i)}).$$

Lemma 4 (Bound on expected excess risk) *The excess risk between the expected ERM population risk $\delta(n, s)$ and the best-in-class population risk $\delta(s)$ is bounded by [18, Proposition 4.1]:*

$$\delta(n, s) - \delta(s) \leq 2 \mathbb{E} \left[\sup_{\theta \in \Theta} |\mathcal{L}_n^{(s)}(h_\theta) - \mathcal{L}_{\text{gen}}^{(s)}(h_\theta)| \right].$$

Proof: Here we directly follow the argument of [18, Proposition 4.1]. Let $\theta_{n,s}^*$ be an ERM and fix any $\theta \in \Theta$. Adding and subtracting the empirical risks $\mathcal{L}_n^{(s)}(h_{\theta_{n,s}^*})$ and $\mathcal{L}_n^{(s)}(h_\theta)$,

$$\begin{aligned} \mathcal{L}_{\text{gen}}^{(s)}(h_{\theta_{n,s}^*}) - \mathcal{L}_{\text{gen}}^{(s)}(h_\theta) &= (\mathcal{L}_{\text{gen}}^{(s)}(h_{\theta_{n,s}^*}) - \mathcal{L}_n^{(s)}(h_{\theta_{n,s}^*})) \\ &\quad + (\mathcal{L}_n^{(s)}(h_{\theta_{n,s}^*}) - \mathcal{L}_n^{(s)}(h_\theta)) \\ &\quad + (\mathcal{L}_n^{(s)}(h_\theta) - \mathcal{L}_{\text{gen}}^{(s)}(h_\theta)). \end{aligned}$$

The middle term is ≤ 0 because $\theta_{n,s}^*$ minimizes the empirical risk, and the first and third are each at most the uniform deviation $\sup_{\theta' \in \Theta} |\mathcal{L}_n^{(s)}(h_{\theta'}) - \mathcal{L}_{\text{gen}}^{(s)}(h_{\theta'})|$. Hence, for every $\theta \in \Theta$,

$$\mathcal{L}_{\text{gen}}^{(s)}(h_{\theta_{n,s}^*}) - \mathcal{L}_{\text{gen}}^{(s)}(h_\theta) \leq 2 \sup_{\theta' \in \Theta} |\mathcal{L}_n^{(s)}(h_{\theta'}) - \mathcal{L}_{\text{gen}}^{(s)}(h_{\theta'})|.$$

Since $\delta(s) = \inf_{\theta \in \Theta} \mathcal{L}_{\text{gen}}^{(s)}(h_\theta)$, we may replace $\mathcal{L}_{\text{gen}}^{(s)}(h_\theta)$ on the left by its infimum $\delta(s)$:

$$\mathcal{L}_{\text{gen}}^{(s)}(h_{\theta_{n,s}^*}) - \delta(s) \leq 2 \sup_{\theta \in \Theta} |\mathcal{L}_n^{(s)}(h_\theta) - \mathcal{L}_{\text{gen}}^{(s)}(h_\theta)|.$$

Taking the expectation yields the claim. $\blacksquare$

Remark 1 On any training example the pointwise difference is at most $2B$, namely $\|F(x) - \hat{F}_\theta(x; O_s)\|_2 \leq \|F(x)\|_2 + \|\hat{F}_\theta(x; O_s)\|_2 \leq 2B$, so integrating over D the per-example loss is bounded by

$$g_\theta(T_s) = \int_D \|F(x) - \hat{F}_\theta(x; O_s)\|_2^2 dx \leq 4B^2|D| \triangleq M.$$

Lemma 5 (Uniform convergence) With finite pseudo-dimension d_s , the pseudo-dimension uniform convergence bound for bounded real-valued classes gives the following [18, Theorem 11.8]: for any $\eta \in (0, 1)$, with probability at least $1 - \eta$, simultaneously for all $\theta \in \Theta$,

$$\mathcal{L}_{\text{gen}}^{(s)}(h_\theta) \leq \mathcal{L}_n^{(s)}(h_\theta) + M \sqrt{\frac{2d_s \log(\frac{en}{d_s})}{n}} + M \sqrt{\frac{\log(1/\eta)}{2n}}.$$

Proof: See the proof of Theorem 11.8 in [18]. $\blacksquare$

Let

$$\varepsilon_n(\eta) \triangleq M \sqrt{\frac{2d_s \log(en/d_s)}{n}} + M \sqrt{\frac{\log(1/\eta)}{2n}}.$$

Then Theorem 5 states that, with probability at least $1 - \eta$, $\mathcal{L}_{\text{gen}}^{(s)}(h_\theta) \leq \mathcal{L}_n^{(s)}(h_\theta) + \varepsilon_n(\eta)$ for all $\theta \in \Theta$. This is one direction of the two-sided bound on $|\mathcal{L}_n^{(s)}(h_\theta) - \mathcal{L}_{\text{gen}}^{(s)}(h_\theta)|$, and the reverse follows by applying the same theorem to the mirrored loss class, as we show next.

Proposition 2 (Two-sided uniform convergence bound)

For any $\eta \in (0, 1)$, with probability at least $1 - \eta$, the following holds simultaneously for all $\theta \in \Theta$:

$$|\mathcal{L}_n^{(s)}(h_\theta) - \mathcal{L}_{\text{gen}}^{(s)}(h_\theta)| \leq \varepsilon_n(\eta/2).$$

Proof: Apply Theorem 5 to the loss class $\mathcal{G}_s$ at confidence level $\eta/2$: with probability at least $1 - \eta/2$,

$$\mathcal{L}_{\text{gen}}^{(s)}(h_\theta) \leq \mathcal{L}_n^{(s)}(h_\theta) + \varepsilon_n(\eta/2), \quad \text{for all } \theta \in \Theta.$$

For the reverse inequality, apply the same theorem to the mirrored class $\{M - g_\theta : \theta \in \Theta\}$, which has the same range $0 \leq M - g_\theta \leq M$ and the same pseudo-dimension d_s (subtracting a constant changes neither). Since $\mathbb{E}[M - g_\theta] = M - \mathcal{L}_{\text{gen}}^{(s)}(h_\theta)$ and its empirical mean is $M - \mathcal{L}_n^{(s)}(h_\theta)$, the theorem gives, with probability at least $1 - \eta/2$,

$$\mathcal{L}_n^{(s)}(h_\theta) \leq \mathcal{L}_{\text{gen}}^{(s)}(h_\theta) + \varepsilon_n(\eta/2) \quad \text{for all } \theta \in \Theta.$$

By the union bound both inequalities hold simultaneously with probability at least $1 - \eta$, giving the two-sided bound. $\blacksquare$

Proposition 3 (Excess-risk bound for ERM) Let $\theta_{n,s}^* \in \arg \min_{\theta \in \Theta} \mathcal{L}_n^{(s)}(h_\theta)$ be an empirical risk minimizer. For any $\eta \in (0, 1)$, with probability at least $1 - \eta$,

$$\mathcal{L}_{\text{gen}}^{(s)}(h_{\theta_{n,s}^*}) - \delta(s) \leq 2\varepsilon_n(\eta/2).$$

Proof: From Proposition 2, the bound $|\mathcal{L}_n^{(s)}(h_\theta) - \mathcal{L}_{\text{gen}}^{(s)}(h_\theta)| \leq \varepsilon_n(\eta/2)$ holds for all θ simultaneously with probability at least $1 - \eta$. Hence, for every $\theta \in \Theta$,

$$\begin{aligned} \mathcal{L}_{\text{gen}}^{(s)}(h_{\theta_{n,s}^*}) &\leq \mathcal{L}_n^{(s)}(h_{\theta_{n,s}^*}) + \varepsilon_n(\eta/2) \\ &\leq \mathcal{L}_n^{(s)}(h_\theta) + \varepsilon_n(\eta/2) \\ &\leq \mathcal{L}_{\text{gen}}^{(s)}(h_\theta) + 2\varepsilon_n(\eta/2), \end{aligned}$$

where the middle step uses that $\theta_{n,s}^*$ minimizes the empirical risk. Since this holds for every θ and $\delta(s) = \inf_{\theta \in \Theta} \mathcal{L}_{\text{gen}}^{(s)}(h_\theta)$, taking the infimum over θ gives $\mathcal{L}_{\text{gen}}^{(s)}(h_{\theta_{n,s}^*}) - \delta(s) \leq 2\varepsilon_n(\eta/2)$ with probability at least $1 - \eta$. $\blacksquare$

Theorem 2 (Estimation error) The expected excess risk of the ERM is controlled, for every $\eta \in (0, 1)$, as

$$\delta(n, s) - \delta(s) = \mathbb{E}[\mathcal{L}_{\text{gen}}^{(s)}(h_{\theta_{n,s}^*})] - \delta(s) \leq 2\varepsilon_n(\eta/2) + \eta M.$$

In particular, choosing $\eta = 1/n$,

$$\begin{aligned} \delta(n, s) - \delta(s) &\leq 2M \sqrt{\frac{2d_s \log(en/d_s)}{n}} \\ &\quad + 2M \sqrt{\frac{\log(2n)}{2n}} + \frac{M}{n}. \end{aligned}$$

Hence, with $M = 4B^2|D|$, for fixed s ,

$$\delta(n, s) - \delta(s) = \mathcal{O}\left(B^2|D| \sqrt{\frac{d_s \log n}{n}}\right) \quad (3)$$

Proof: Fix $\eta \in (0, 1)$ and define the excess-risk random variable $\Delta_n \triangleq \mathcal{L}_{\text{gen}}^{(s)}(h_{\theta_{n,s}^*}) - \delta(s)$. By Proposition 3, with probability at least $1 - \eta$, $\Delta_n \leq 2\varepsilon_n(\eta/2)$. Also, since $0 \leq g_\theta \leq M$ for every θ , we have $0 \leq \delta(s) \leq M$ and $0 \leq \mathcal{L}_{\text{gen}}^{(s)}(h_{\theta_{n,s}^*}) \leq M$. Together with $\mathcal{L}_{\text{gen}}^{(s)}(h_{\theta_{n,s}^*}) \geq \delta(s)$ this gives $0 \leq \Delta_n \leq M$. Therefore

$$\begin{aligned} \mathbb{E}[\Delta_n] &\leq 2\varepsilon_n(\eta/2) \mathbb{P}(\Delta_n \leq 2\varepsilon_n(\eta/2)) \\ &\quad + M \mathbb{P}(\Delta_n > 2\varepsilon_n(\eta/2)) \\ &\leq 2\varepsilon_n(\eta/2) + \eta M. \end{aligned}$$

Since $\mathbb{E}[\Delta_n] = \delta(n, s) - \delta(s)$, choosing $\eta = 1/n$ gives the stated bound, and hence, for fixed s , $\delta(n, s) - \delta(s) = \mathcal{O}(B^2|D| \sqrt{d_s \log n/n})$. $\blacksquare$

Corollary 2 (Total reconstruction error) Under the stated assumptions, there exist constants $C_1, C_2 > 0$ such that

$$\delta(n, s) \leq C_1 s^{-2/d} + C_2 \sqrt{\frac{d_s \log n}{n}},$$

up to lower-order terms in n .

The first term is the approximation error, bounded in (2), which is caused by observing only s samples of each radio map. The second term is the estimation error, bounded in (3), due to learning the reconstructor using a finite dataset of n examples. For a two-dimensional domain $d = 2$, the approximation bound becomes $\mathcal{O}(1/s)$. Thus, under the ideal assumptions, the approximation term has an inverse dependence on the number of observed samples per map in two dimensions.

In practice, the assumptions underlying this rate may only be approximately satisfied in RME settings. For example, sampling may be non-uniform since measurements are often unavailable inside buildings, and radio maps may violate smoothness assumptions due to sharp transitions at building boundaries and shadowing effects. Whether this rate describes how practical reconstruction error decays when these assumptions are relaxed is an empirical question, which we study next.

V. MODEL ARCHITECTURE

In this section, we propose RadioET, an energy-based reconstruction model based on ConvNeXt layers [19] and the Energy Transformer [15]. Both the encoder and decoder use hierarchical convolutional layers. The encoder extracts local features and reduces the spatial resolution of the input at each stage. At the bottleneck, Energy Transformer blocks model global spatial relationships and estimate missing regions from sparse observations. The decoder reconstructs the output by combining encoder features through skip connections. Fig. 1 shows the overall architecture.

The Energy Transformer [15] uses a single recurrent block that iteratively updates its tokens to minimize a global energy term. Let $x_C \in \mathbb{R}^D$ be the token at spatial location C and g_C its layer-normalized version. Throughout, B, C index the N tokens and $j = 1, \dots, D$ runs over feature coordinates. The block defines two separate energy terms. The first is the attention energy,

$$E^{\text{ATT}} = -\frac{1}{\beta} \sum_{h=1}^H \sum_C \log \left(\sum_{B \neq C} \exp(\beta A_{hBC}) \right), \quad (4)$$

where the score $A_{hBC} = \sum_a K_{ahB} Q_{ahC}$ couples tokens B and C within attention head h , built from keys $K_{ahB} = \sum_j W_{ahj}^K g_{jB}$ and queries $Q_{ahC} = \sum_j W_{ahj}^Q g_{jC}$. Here W^K, W^Q are learnable projections, a indexes the per-head attention channels, and β is an inverse temperature [15].

The second is the Hopfield (associative-memory) energy that pulls each token toward plausible patterns,

$$E^{\text{HN}} = -\sum_B \sum_{\mu=1}^M G \left(\sum_j \xi_{\mu j} g_{jB} \right), \quad (5)$$

where the M rows of ξ are learnable memory patterns and G is a smooth nonlinearity whose derivative $G' = r$ is the activation function. The total global energy at time step t , denoted E^t , is the sum of the terms given in (4) and (5). The

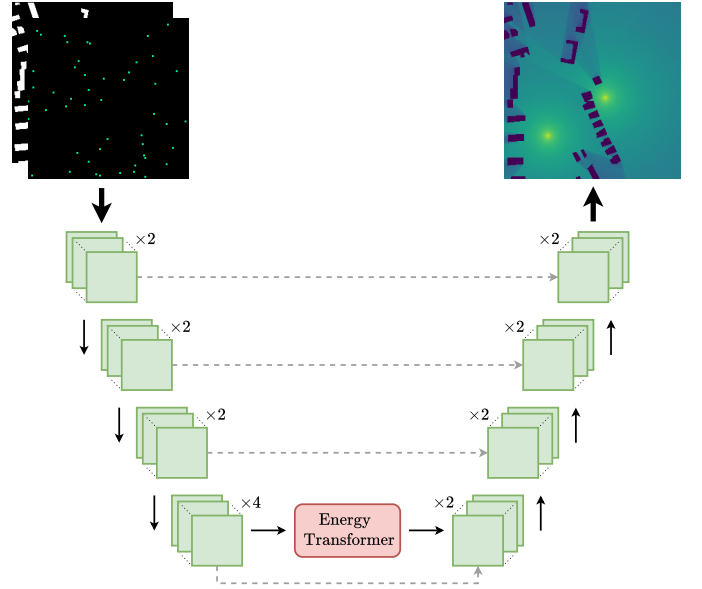


Fig. 1. Architecture of RadioET. Dashed lines indicate skip connections.

tokens are evolved by gradient descent on this energy for a total of T steps with step size α [15]:

$$x^{t+1} = x^t - \alpha \nabla_g E^t, \quad t = 1, \dots, T.$$

Lastly, the choices of architectural parameters we employed for RadioET are given in Table I.

TABLE I
RADIOET ARCHITECTURAL PARAMETERS

Parameter	Value	Parameter	Value
Stage dims	48, 96, 192, 384	Number of heads	8
Encoder blocks	2, 2, 2, 4	Hopfield r	Softmax
Decoder blocks	2, 2, 2, 2	Attention β	$\frac{1}{\sqrt{64}}$
Bottleneck dim	512	Time steps T	12
QK dim	64	Step size α	0.1

VI. EXPERIMENTS

We conduct two main experiments using the RadioMapSeer dataset [20]. To expand the sampling space and introduce greater stochastic variation, we construct a multi-transmitter RME benchmark.

A. Baseline Comparisons

In order to create a realistic and challenging multi-transmitter RME benchmark, we superpose pathloss fields from the RadioMapSeer dataset [20] in linear power space. Specifically, we randomize the number, locations, and powers of the transmitters. Let $\mathcal{N}_{\text{tx}}$ denote the set of possible transmitter counts and let $\mathcal{M}_{\text{tx}}$ denote the transmit power-offset range in dB. Because weaker signals are truncated to -147 dB in the original dataset, sentinel pixels at this floor value contribute no linear power to the mixture. After synthesis, all outputs are placed on a common path loss scale

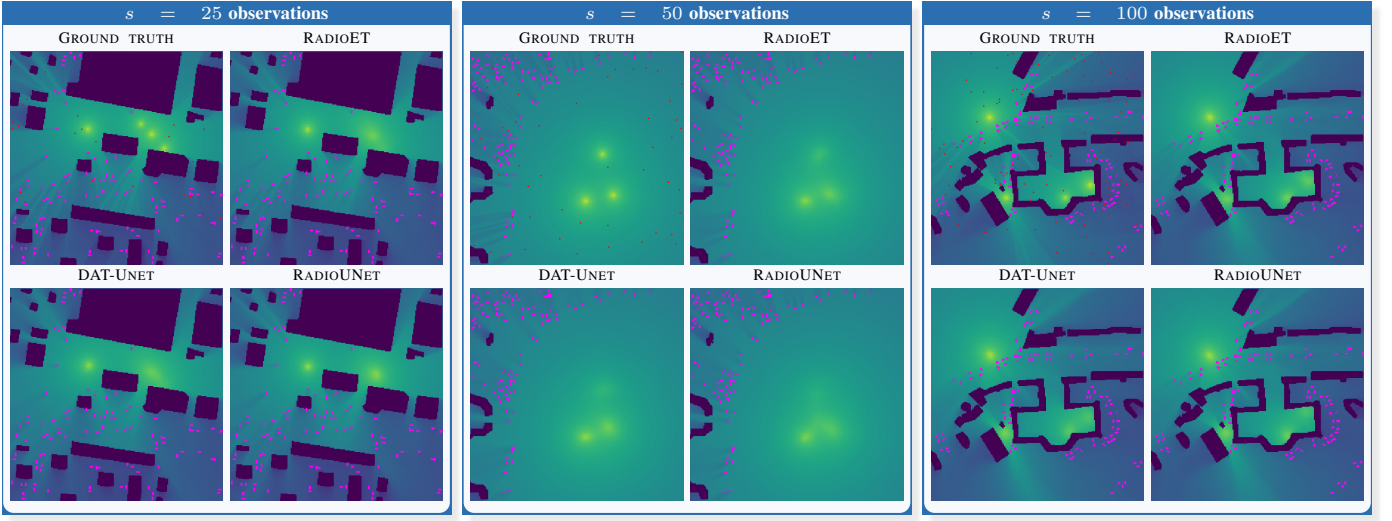


Fig. 2. Qualitative comparisons for multi-transmitter RME at $s = 25, 50$, and 100 observed samples (red points). Purple and pink blocks denote buildings and cars, respectively.

Algorithm 1 Multi-transmitter map synthesis

Input: Single-TX path loss maps $\{L_k\}_{k=1}^{80}$ in dB, count support $\mathcal{N}_{\text{tx}}$, power-offset range $\mathcal{M}_{\text{tx}}$

- 1: $n \sim \mathcal{U}(\mathcal{N}_{\text{tx}})$ {number of active transmitters}
- 2: sample distinct indices $\mathcal{I} = \{i_1, \dots, i_n\}$ uniformly from $\{1, \dots, 80\}$
- 3: $T(x) \leftarrow 0$ for all pixels x
- 4: **for** each $i \in \mathcal{I}$ **do**
- 5: $m_i \sim \mathcal{U}(\mathcal{M}_{\text{tx}})$ {power offset in dB}
- 6: $\alpha_i \leftarrow 10^{m_i/10}$ {linear power scale}
- 7: $M_i(x) \leftarrow \mathbf{1}_{\{L_i(x) > -147\}}$
- 8: $T(x) \leftarrow T(x) + \alpha_i M_i(x) 10^{L_i(x)/10}$
- 9: **end for**
- 10: $L(x) \leftarrow \begin{cases} \max\{10 \log_{10} T(x), -147\}, & T(x) > 0, \\ -147, & T(x) = 0 \end{cases}$
- 11: **return** L

$[0, 1]$, where the lower end corresponds to the RadioMapSeer floor of -147 dB and the upper end corresponds to the worst-case aggregate value of $-47 + \max \mathcal{M}_{\text{tx}} + 10 \log_{10}(\max \mathcal{N}_{\text{tx}})$. Using the procedure outlined in Algorithm 1, we generate random training examples on the fly during training.

We create dataset splits by randomly assigning 501, 100, and 100 city layouts to the training, validation, and test sets, respectively. We use the radio maps generated by DPM simulations from the RadioMapSeer dataset. In this experiment, we vary the number of observed samples (s) uniformly within $[10, 100]$ during training to evaluate the models' flexibility under varying degrees of sparsity. This corresponds to sampling between 0.015% and 0.15% of the spatial domain. We also include the car objects in the inputs to introduce additional obstructions.

We compare RadioET against DAT-Unet [3] and RadioUNet [7], which we reimplemented in PyTorch Lightning

to ensure identical training pipelines. During validation and evaluation phases, we calculate RMSE and NMSE only on unobserved free-space pixels. For PSNR and SSIM, predictions are overwritten with ground truth values at observed pixels and zeros inside buildings. For evaluation metrics, we clamp predictions to $[0, 1]$.

All models use the same training setup given in Table III. Since RadioUNet consists of two training stages, we train both stages for 150 epochs. Additionally, we perform a lightweight hyperparameter search on the validation set. We perform a grid search over learning rates. For DAT-Unet, we additionally tune the drop path rate. Specifically, DAT-Unet is evaluated on the learning rates $\{5 \times 10^{-4}, 1 \times 10^{-3}, 2 \times 10^{-3}\}$ and drop path rates $\{0.0, 0.1, 0.2\}$. RadioET is evaluated using learning rates $\{5 \times 10^{-4}, 1 \times 10^{-3}, 2 \times 10^{-3}\}$ and RadioUNet using learning rates $\{1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}, 2 \times 10^{-3}\}$. All models are trained with a masked MSE loss computed only on unobserved and free-space pixels. The inputs to the models consist of the observed path loss values and the binary observation, building, and car masks.

We train each model using 3 random seeds and report the mean and standard deviation of the evaluation metrics averaged on the test set. The results under varying sparsity regimes are summarized in Table II. In all metrics and sparsity regimes, the performance gains of RadioET are consistent. We also provide qualitative comparisons in Fig. 2.

B. Empirical sparsity scaling

We study how reconstruction error scales with the number of observed samples and compare the empirical trend to the rate $\delta(s) = \mathcal{O}(1/s)$ in Section IV. Separate RadioET models are trained at sparsities $s \in \{10, 20, 40, 80, 160\}$ with a single seed since variance is small (Table II). To better align with the reconstruction loss used in our analysis, MSE is computed over all pixels. The number of training examples n is kept fixed through the number of city layouts in the training split,

TABLE II
RESULTS ON THE TEST-SET. **BOLD MARKS THE BEST METHOD.**

$s \in$	Model	RMSE ↓	NMSE ↓	SSIM ↑	PSNR ↑
[10, 25]	DAT-Unet	0.03814 ±	0.00814 ±	0.9553 ±	30.55 ±
		0.00013	0.00006	0.0001	0.03
	RadioUNet	0.03862 ±	0.00834 ±	0.9588 ±	30.59 ±
		0.00016	0.00007	0.0002	0.04
	RadioET	0.03652 ±	0.00746 ±	0.9599 ±	31.06 ±
		0.00046	0.00019	0.0009	0.14
[25, 50]	DAT-Unet	0.02700 ±	0.00408 ±	0.9623 ±	33.17 ±
		0.00006	0.00002	0.0001	0.02
	RadioUNet	0.02635 ±	0.00388 ±	0.9666 ±	33.54 ±
		0.00017	0.00005	0.0003	0.07
	RadioET	0.02438 ±	0.00333 ±	0.9684 ±	34.16 ±
		0.00052	0.00014	0.0011	0.21
[50, 100]	DAT-Unet	0.02223 ±	0.00277 ±	0.9657 ±	34.62 ±
		0.00006	0.00001	0.0000	0.02
	RadioUNet	0.02060 ±	0.00238 ±	0.9707 ±	35.36 ±
		0.0002	0.00005	0.0003	0.09
	RadioET	0.01923 ±	0.00207 ±	0.9726 ±	35.93 ±
		0.0005	0.00011	0.0011	0.25
[10, 100]	DAT-Unet	0.02695 ±	0.00406 ±	0.9630 ±	33.52 ±
		0.00005	0.00002	0.0001	0.02
	RadioUNet	0.02612 ±	0.00382 ±	0.9675 ±	34.04 ±
		0.00017	0.00005	0.0003	0.08
	RadioET	0.02446 ±	0.00335 ±	0.9693 ±	34.61 ±
		0.00046	0.00013	0.0011	0.22

TABLE III
TRAINING SETUP

Parameter	Value	Parameter	Value
Optimizer	AdamW	Batch Size	128
β_1, β_2	0.9, 0.95	Epochs	150
Weight Decay	10^{-4}	$\mathcal{N}_{\text{tx}}$	{2, 3, 4}
LR Schedule	Cosine	$\mathcal{M}_{\text{tx}}$	(−6, 0) dB
Parameter	DAT-Unet	RadioUNet	RadioET
Learning rate	1×10^{-3}	5×10^{-4}	2×10^{-3}
DropPath Rate	0.1	–	–

while Algorithm 1 generates different multi-transmitter maps. This isolates the effect of the number of observed samples per map.

Since $\delta(s)$ is a population best-in-class risk, it cannot be measured directly from a finite dataset, so we instead bracket it. Let θ^* attain the infimum in $\delta(s)$. Because the ERM minimizes the empirical risk, $\mathcal{L}_n^{(s)}(h_{\theta_{n,s}^*}) \leq \mathcal{L}_n^{(s)}(h_{\theta^*})$ for every training set. Since θ^* is fixed, its expected empirical

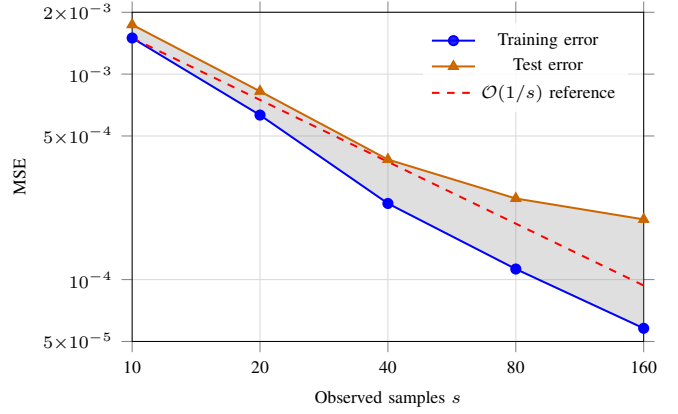


Fig. 3. Empirical sparsity scaling. Training and test MSE bracket the best-in-class population risk $\delta(s)$ (shaded band). The observed decay is consistent with the $\mathcal{O}(1/s)$ reference rate.

risk equals $\delta(s)$. Taking expectations therefore yields the lower bound $\mathbb{E}[\mathcal{L}_n^{(s)}(h_{\theta_{n,s}^*})] \leq \delta(s)$. Also, $\delta(s) \leq \delta(n, s)$ because no data-dependent $\theta_{n,s}^*$ can attain population risk below the infimum. Combining, $\mathbb{E}[\mathcal{L}_n^{(s)}(h_{\theta_{n,s}^*})] \leq \delta(s) \leq \delta(n, s)$, so the training and test errors of the ERM bracket $\delta(s)$.

Fig. 3 shows the resulting sparsity scaling on logarithmic axes. The horizontal axis is the number of observed samples s , and the vertical axis is the full-map reconstruction MSE. The blue and orange curves show the training and test errors, respectively, and the shaded region between them gives the empirical bracket for the best-in-class population risk $\delta(s)$.

The red dashed curve shows the $\mathcal{O}(1/s)$ reference predicted by the approximation term in Corollary 2 (with $d = 2$). Even though the assumptions underlying the bound are only approximately satisfied in practice, the shaded region remains consistent with this reference over the range of s we examine. The remaining gap between training and test errors reflects finite-sample generalization effects. In particular, the training error measures the empirical risk minimized on the finite training set (up to optimization error), whereas the test error measures performance on unseen radio maps generated from the same benchmark distribution. As s increases, the approximation error decreases, but the finite training set can still limit how closely the learned reconstructor matches the best-in-class population risk, which explains why the train–test bracket remains visible.

VII. DISCUSSION AND CONCLUSION

We formulated sampling-based multi-transmitter RME as a sparse reconstruction problem and compared our empirical findings to a reference bound on the expected reconstruction risk. Empirically, RadioET achieves strong reconstruction performance on the multi-transmitter benchmark, and the error-decay experiments show monotone improvement with increasing s . The empirical decay is consistent with $\mathcal{O}(1/s)$, and establishing the rate under the relaxed assumptions of practical RME remains open.

While RadioET achieves notable gains in accuracy, these improvements introduce a trade-off with computational com-

plexity. The energy-based block utilizes T iterative update steps per forward pass, and processing convolutional layers at high resolutions requires additional computational resources. Specifically, full training takes approximately 8h20m versus 2h40m for DAT-Unet and 2h10m for RadioUNet on $4\times$ RTX 6000 Ada GPUs under the identical pipeline.

REFERENCES

- [1] D. Romero and S.-J. Kim, "Radio map estimation: A data-driven approach to spectrum cartography," *IEEE Signal Processing Magazine*, vol. 39, no. 6, pp. 53–72, 2022.
- [2] X. Wang, Y. Pan, and N. Cheng, "A tutorial on learning-based radio map construction: Data, paradigms, and physics-awareness," 2026. [Online]. Available: <https://arxiv.org/abs/2603.17499>
- [3] K. Liu, C. Qiu, K. Chen, Q. Zheng, L. Song, and Y. Wang, "Paying deformable attention to sparse spatial observations for deep radio map estimation," *IEEE Transactions on Cognitive Communications and Networking*, vol. 12, pp. 1436–1450, 2026.
- [4] E. Koyuncu, "Performance gains of optimal antenna deployment in massive MIMO systems," *IEEE Transactions on Wireless Communications*, vol. 17, no. 4, pp. 2633–2644, Apr. 2018.
- [5] E. Koyuncu, M. Shabanighazikelayeh, and H. Seferoglu, "Deployment and trajectory optimization of UAVs: A quantization theory approach," *IEEE Trans Wireless Commun.*, Dec. 2018.
- [6] D. Denkovski, V. Atanasovski, L. Gavrilovska, J. Riihijärvi, and P. Mähönen, "Reliability of a radio environment map: Case of spatial interpolation techniques," in *CROWNCOM*, 2012.
- [7] R. Levie, Ç. Yapar, G. Kutyniok, and G. Caire, "RadioUNet: Fast radio map estimation with convolutional neural networks," *IEEE Transactions on Wireless Communications*, vol. 20, no. 6, pp. 4001–4015, 2021.
- [8] H. Jia, N. Cheng, X. Wang, C. Zhou, R. Sun, and X. Shen, "Radiomamba: Breaking the accuracy-efficiency trade-off in radio map construction via a hybrid mamba-unet," *IEEE Trans. on Network Science and Engineering*, vol. 13, pp. 2454–2468, 2026.
- [9] X. Wang, K. Tao, N. Cheng, Z. Yin, Z. Li, Y. Zhang, and X. Shen, "Radiodiff: An effective generative diffusion model for sampling-free dynamic radio map construction," *IEEE Transactions on Cognitive Communications and Networking*, vol. 11, no. 2, pp. 738–750, 2025.
- [10] H. Jia, W. Chen, Z. Huang, L. Wang, H. Xiao, N. Jia, K. Wu, S. Lai, B. Tian, and Y. Yue, "Physics-informed representation alignment for sparse radio-map reconstruction," in *Proceedings of the 33rd ACM International Conference on Multimedia*, 2025.
- [11] S. Zhang, A. Wijesinghe, and Z. Ding, "Rme-gan: A learning framework for radio map estimation based on conditional generative adversarial network," *IEEE Internet of Things Journal*, vol. 10, no. 20, pp. 18 016–18 027, 2023.
- [12] P. Zeng and J. Chen, "Uav-aided joint radio map and 3d environment reconstruction using deep learning approaches," in *ICC 2022 - IEEE International Conference on Communications*, 2022, pp. 5341–5346.
- [13] T. Chen, Z. Zhou, Z. Fang, W. Zou, K. Liu, K. Chen, Y. Zhang, and Y. Wang, "Radiodun: A physics-inspired deep unfolding network for radio map estimation," *IEEE Transactions on Cognitive Communications and Networking*, 2026.
- [14] B. Oral and E. Koyuncu, "REPA: Random-order embedding predictive autoregression for sparse vector field reconstruction," in *IEEE International Conference on Image Processing*, 2026.
- [15] B. Hoover, Y. Liang, B. Pham, R. Panda, H. Strobelt, D. H. Chau, M. Zaki, and D. Krotov, "Energy transformer," in *Advances in Neural Information Processing Systems*, 2023.
- [16] Q. Zhang, D. Krotov, and G. E. Karniadakis, "Operator learning for reconstructing flow fields from sparse measurements: An energy transformer approach," *J. Comp. Physics*, 2025.
- [17] G. Biau and L. Devroye, *Lectures on the Nearest Neighbor Method*. Springer International Publishing, 2015.
- [18] M. Mohri, A. Rostamizadeh, and A. Talwalkar, *Foundations of Machine Learning*, 2nd ed. MIT Press, 2018.
- [19] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A convnet for the 2020s," in *CVPR*, 2022.
- [20] Ç. Yapar, R. Levie, G. Kutyniok, and G. Caire, "Dataset of pathloss and toa radio maps with localization application," 2024. [Online]. Available: <https://arxiv.org/abs/2212.11777>